

Chien-Hsien Chu

FULL-STACK • DEVOPS • PLATFORM ENGINEER • 5+ YEARS WORK EXPERIENCE

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Education

National Yang Ming Chiao Tung University (NYCU)

Hsinchu, Taiwan

M.S. IN COMPUTER SCIENCE

Sep. 2023 – Sep. 2026

- Overall GPA: 4.2/4.3 · Advisor: Prof. Chien-Chao Tseng
- Thesis: *Observability-Driven Root-Cause Analysis and Recovery Framework for Cloud-Native 5G Core Network Functions.*

Work Experience

ISCOM Online International Information Inc.

Taichung, Taiwan

R&D ENGINEER, R&D DEPARTMENT

Jul. 2022 – Aug. 2023

- Established a Unified Frontend Framework with **Next.js**, adopted across multiple internal products.
- Designed and implemented internal **GitLab CI/CD Pipelines** to automate build, test, and deployment workflows.
- Built and operated production **Kubernetes** clusters across on-premises and cloud environments (**AKS**).

FULL-STACK ENGINEER / TECHNICAL PROJECT MANAGER, BUSINESS UNIT

Jul. 2018 – Jun. 2022

- Architected and delivered over **45 web and enterprise projects**: E-Commerce, POS, and GIS-Based Systems.
- Built full-stack solutions with **ASP.NET Core**, **Laravel**, **Next.js**, and **Vue.js**, backed by **MSSQL** and **MySQL** databases.
- Integrated **Payment Gateways**, **LINE Official Account**, and **POS Hardware APIs** into client systems.
- Modernized legacy **.NET Framework** codebase to cross-platform, containerizable **.NET Core** architecture.
- Supported clients on **Azure Cloud** deployment, including Web Apps, VMs, and managed databases.

Internship

Taichung, Taiwan

SOFTWARE ENGINEER INTERN

Jul. 2015 – Jun. 2018

- Three successive internships at IT companies: **ISCOM Inc.**, **BitCloud Co.**, and **RCE Power Inc.**
- Web application development, front-end performance work, and internal data-collection automation.

Academic & Technical Experience

Observability-Driven Root-Cause Analysis and Recovery Framework for Cloud-Native 5G Core Network Functions

M.S. Thesis

GRADUATE RESEARCH ASSISTANT, WIRELESS & INTERNET LAB (WINLAB), NYCU

Feb. 2026 – Sep. 2026

- Built an End-to-End Observability and Auto-Remediation Platform for multi-cluster **Kubernetes**: Non-Intrusive Telemetry Collection, Generation, and Correlation, Fault Detection, Root-Cause Analysis, and Guarded Remediation.
- Telemetry Collection, Generation, and Correlation: zero-code trace and RED Metric generation via **eBPF (Grafana Beyla)**, shipped over **OTLP** by **OpenTelemetry Collectors**, enriched to one correlation key.
- Telemetry Storage and Fault Detection: **Prometheus**, **Loki**, **Tempo**, **Alertmanager**.
- LLM-Based RCA Agent: Agentic Investigation Loop over live telemetry, guided by injected environment knowledge.
- RAG-Based Recovery Agent: Knowledge Base retrieval, then **Auto / Approval / Escalation** remediation over **MCP**.
- Transparent State-Recovery Proxy: record-and-replay recovery of a stateful service's in-memory state after a restart.

🔗 Infrastructure Automation & Management for Laboratory Environment

📄 source code

SYSTEM ADMINISTRATOR, WIRELESS & INTERNET LAB (WINLAB), NYCU

Jul. 2024 – Sep. 2026

- Independently migrated a Kubernetes platform from **Kubespray + Terraform** to an immutable, GitOps Architecture using **Talos Linux** and **Flux CD**, including cluster bootstrap, node lifecycle automation, and PV-level data migration.
- Executed multi-application, multi-version upgrades involving rollbackable database upgrades and data migrations.
- Integrated **Velero**, **MinIO**, **Grafana Stack**, and **Keycloak** for backup, object storage, observability, and access control.
- Managed **Harvester** clusters and migrated to **Proxmox VE** for improved stability and flexibility.

System Architecture Planning and Azure Infrastructure Consultation

TECHNICAL CONSULTANT, KOI STUDIO LTD. Dec. 2023 – Nov. 2024

- Advised on system architecture and **Azure** infrastructure design for a software product company.
- Resolved deployment and integration issues between cloud services and external vendor APIs.

Kubernetes Infrastructure and DevOps Workflow Development

TECHNICAL ASSISTANT, INFORMATION TECHNOLOGY SERVICE CENTER, NYCU Sep. 2023 – Aug. 2024

- Built **Kubernetes** infrastructure for institutional systems.
- Enhanced CI/CD automation using **GitLab CI** to improve delivery reliability.

Kubernetes Networking and Security Enhancement for Department Infrastructure

SYSTEM TEACHING ASSISTANT, CSIT, NYCU Jul. 2024 – Sep. 2026

- Migrated networking from **Flannel** to **Calico** for fine-grained **NetworkPolicy** enforcement.
- Implemented **Istio Egress Gateway** for unified outbound traffic management.
- Evaluated **External Secrets Operator (ESO)** and **OpenBao (Vault fork)** for Kubernetes secret management.

Multi-Cluster Management System

GRADUATE STUDENT ASSISTANT, WIRELESS & INTERNET LAB (WINLAB), NYCU Sep. 2023 – Jan. 2024

- **Multi-Cluster Management System** integrating **Karmada**, **Cilium**, and **Thanos** for orchestration and observability.

Technical Skills

Cloud & Container Orchestration	Kubernetes (AKS, Talos Linux, Kubespray), Docker, Helm, Flux CD, Harbor, Karmada
Observability	Prometheus, Grafana, Loki, Tempo, Thanos, OpenTelemetry (OTLP), Alertmanager
AI & LLM Systems	LLM Agents, RAG, MCP, Prompt Engineering
Networking & Systems	Linux System Administration, IPv4 Networking, eBPF, MetalLB, Traefik, Istio, Calico, Cilium
Infrastructure & Automation	Proxmox VE, Harvester, Terraform (OpenTofu), Ansible, Velero, GitLab CI/CD, GitOps
Security & Identity	Keycloak, HashiCorp Vault (OpenBao), NetworkPolicy, Cert-Manager, OAuth2 / OIDC
Programming & Scripting	Bash, Python, Go, C++, C# (.NET MVC/Core), PHP (Laravel), JavaScript (Vue.js / Next.js), SQL (MSSQL / MySQL)

Awards & Honors

Academic Excellence Award, National Yang Ming Chiao Tung University (NYCU)	Spring 2024
Academic Excellence Award, National Yang Ming Chiao Tung University (NYCU)	Fall 2023